

File 69: Georgia EPD Trout Stream Buffer Variance Application

State Permitting and Sediment Control inside the 25-Foot Trout Stream Buffer

State Permitting & Soil Erosion Standard: This document represents the official Georgia DNR Environmental Protection Division (EPD) Trout Stream Buffer Variance Application narrative for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It outlines compliance with O.C.G.A. § 12-7-6(b)(15), detailing the sediment bypass pump-around schematics, turbidity barrier specifications, and bank vegetative stabilization schedules inside the 25-foot protective buffer.

I. Applicant & Project Identification

- **Applicant:** Blue Ridge Stream Restoration & Mitigation LLC (on behalf of landowner Roya Morris)
- **Permit Authority:** Georgia Department of Natural Resources, Environmental Protection Division (EPD)
- **EPD District Office:** Mountain District Office (Cartersville, GA)
- **Regulatory Reference:** Variance for land-disturbing activities within the 25-foot state-mandated buffer of a designated coldwater trout stream (O.C.G.A. § 12-7-6(b)(15)).
- **Soil & Water Conservation Commission (GSWCC) Plan Number:** 2026-FANNIN-003 (Red Card certified plan).

II. Project Narrative & Buffer Encroachment Justification

Description of Proposed Activity

The applicant proposes to perform fluvial geomorphic stream restoration along 1,500 linear feet of degraded pasture channel on Anderson Creek. Instream operations require heavy excavation equipment (excavators, loaders) to enter and work within the state-mandated 25-foot buffer to grade vertical, collapsing clay banks to a stable 3:1 profile, install geomorphic rock/log structures, and plant a dense native forest canopy.

Why Encroachment is Unavoidable

To stabilize the eroding pasture banks and restore a stable stream channel, heavy machinery must directly access the stream bed and bankful elevations. It is physically impossible to stabilize lateral bank failures, reconstruct natural meander curves, or place heavy boulder/log structures (rock cross-vanes, toe-wood complexes) without working directly within the 25-foot protective buffer.

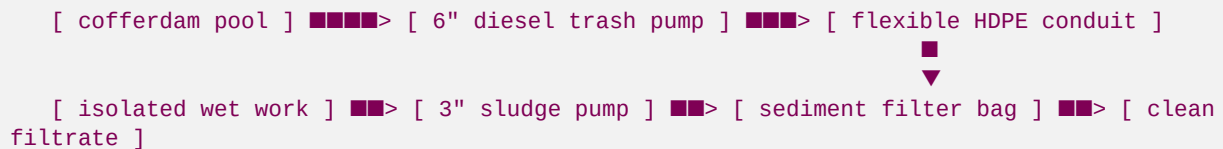
Net Environmental Benefit Narrative

Although this project causes temporary disturbance within the 25-foot buffer, it will result in a permanent, massive reduction in regional erosion. Active lateral pasture banks currently dump **286 tons of sediment per year** directly into the creek, choking the spawning gravels of wild **Brown Trout (*Salmo trutta*)**. Grading these banks back to a stable 3:1 slope, wrapping them immediately in double-net coir matting, and installing a high-density native forest buffer will reduce sediment loading by **98%**, permanently improving water quality.

III. Fluvial Erosion & Sedimentation Control Plan

To ensure that land-disturbing activities within the 25-foot buffer do not release sediment downstream or violate Georgia's 25-NTU water quality standard, we will implement this robust **Fluvial Erosion & Sedimentation Control Plan**:

EPD SEDIMENT CONDUIT & FILTER DETAIL



1. High-Velocity Clean Water Pump-Around (GSWCC detail: Bypass-1)

- **Dam & Pump Configuration:** At the upstream limit of each 100-foot construction zone, construct a temporary cofferdam using clean washed gravel covered in heavy-duty 10-mil plastic sheeting. Place a 6" diesel-driven high-performance trash pump at the dam pool.
- **Bypass Conduit:** Route clean, unpolluted stream water through a flexible 6" HDPE pipeline around the active work zone, discharging it downstream onto a splash pad made of Class II rip-rap to prevent scour.
- **Monitoring Standard:** The bypass system must operate continuously during active instream excavation, keeping the work area completely isolated from the active streamflow.

2. Work Area Seepage Silt Filtration (GSWCC detail: Sediment Bag-2)

- **Seepage Ingress:** Active groundwater seepage and minor bank seepage will pool in the dry work channel.
- **Filter Bag Siting:** Place a 3" high-sludge pump in the collection pool and route muddy water into a heavy-duty 15' x 15' non-woven geotextile **Sediment Filter Bag** placed on a flat, grassed upland area at least 30 feet outside the riparian buffer.
- **Filtration Action:** The filter bag traps all suspended clay, silt, and sand particles. Clean, clear filtrate water flows out of the bag's pores and sheet-flows over grass back to the stream, ensuring zero sediment reaches the creek.

3. Double Sediment Barriers (GSWCC detail: Sd1-C)

- Install two parallel lines of high-sensitivity Type C (tensile-strength reinforced) silt fencing along the entire landward boundary of the 25-foot work buffer, preventing any upland overland runoff from carrying silt into the stream.

IV. Vegetative Buffer Stabilization & Planting Schedule

Upon completing bank grading, all disturbed areas within the 25-foot buffer must be stabilized geomechanically and vegetatively within **24 hours** using this dense planting matrix:

1. Geomechanical Bank Stabilization

Grade bank slopes back to a stable 3:1 slope, apply a native riparian seed mix (Switchgrass, Virginia Wildrye, Fox Sedge) at a rate of 25 lbs/acre, and cover immediately with **700g/sqm double-net Coir Fiber Matting (natural coconut husk matrix)**. Secure the coir matting tightly using 12" wooden stakes driven in a diamond pattern at 2-foot intervals.

2. Bioengineered Live Staking Schedule

Insert dormant, live vegetative cuttings through the coir matting directly into the bankfull elevations at a density of **1 live stake per 2 square feet**:

- **Black Willow (Salix nigra)**: 900 live stakes (3' length, 1" diameter). Black willow roots rapidly, binding bank soils together.
- **Silky Dogwood (Cornus amomum)**: 900 live stakes (3' length, 1" diameter). Provides excellent root network and bank structural stability.

3. Riparian Shading Tree Canopy Planting Schedule

Plant bare-root, high-status native trees within the 25-foot state buffer at a density of **400 trees/acre** to restore permanent coldwater shading for the brown trout spawning runs:

Native Tree Species	Siting Placement Zone	Root Type	Shading Canopy Benefit
Eastern Hemlock (Tsuga canadensis)	Bank bankfull toe	Deep coniferous	High-density evergreen year-round shading
Tag Alder (Alnus serrulata)	Instream bank water line	Dense fibrous	Nitrogen-fixing organic inputs for insects
River Birch (Betula nigra)	Mid-bank slope	Deep lateral	Fast-growing canopy, locks bank soils
Rosebay Rhododendron (R. maximum)	Buffer understory	Dense matting	High-density solar heat attenuation (LAI >= 5.8)

Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.